Local Spectral Closure of the Capacity–Metric Bridge

LC-O2-O1 of the Lorentz–Capacity Sub-Programme

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2026

Abstract

We close the local bridge between the capacity radius and the temporal co-metric component in the Lorentz-capacity formulation of the Cosmochrony programme. The local theory introduces an occupancy field $\epsilon(x)$ through the residual Born–Infeld capacity radius $R(x)^2 = 1 - \epsilon(x)$, while the metric formulation introduces the same radius through $R(x)^2 = -2g^{\tau\tau}(x)$. Their equivalence was previously stated as an open analytic bridge in [2]. We prove that the identity

$$\epsilon(x) = 1 + 2g^{\tau\tau}(x)$$

follows directly from local admissibility data. The key input is that $\epsilon(x)$ is a quadratic occupancy in the rank-three sector $\mathrm{Sym}^2(V_\rho)$, normalised by the maximal Born–Infeld capacity $\sigma_{\max} = c_\chi^2$. A local version of the admissibility-weight argument of [4] then gives

$$A_\tau(x) = 2\big(1 - \epsilon(x)\big).$$

Using the principal-symbol identification of the effective operator [1, 3], together with the flat-space normalisation $A_\tau^\infty = 2$, $g^{\tau\tau,\infty} = -\frac{1}{2}$, yields

$$g^{\tau\tau}(x) = -\frac{A_\tau(x)}{4}.$$

Therefore $R(x)^2 = 1-\epsilon(x) = -2g^{\tau\tau}(x)$. No use is made of the Einstein equation. The bridge is closed spectrally from the local admissibility form alone.

Keywords. local capacity radius; spectral occupancy; admissibility-weight reduction; temporal co-metric; Born–Infeld capacity; effective Lorentzian geometry; gravitational time dilation; Cosmochrony.

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1 Introduction

The Lorentz-capacity formulation [2] introduces a local occupancy field $\epsilon(x)$ measuring how much of the projective capacity at x is already locked by admissible spectral structure. On the capacity side, the residual radius is defined by

$$R(x)^2 := 1 - \epsilon(x).$$

On the metric side, the same quantity is defined by the temporal co-metric component,

$$R(x)^2 := -2g^{\tau\tau}(x).$$

The open problem LC-O2-O1 of [2] (Remark 3.2 and Section 7.3) is to prove

$$1 - \epsilon(x) = -2g^{\tau\tau}(x)$$

from admissibility data alone, without importing the Einstein equation from the spectral gravity branch.

The natural object controlling this bridge is the local temporal coefficient $A_\tau(x)$ of the effective admissibility operator L_{eff}. In the homogeneous flat limit, this coefficient is fixed by [3]:

$$A_\tau^\infty = 2.$$

When a fraction $\epsilon(x)$ of the local admissible fibre is already occupied, the coefficient is reduced by the free quadratic capacity fraction:

$$A_\tau(x) = 2\big(1 - \epsilon(x)\big).$$

This is the local counterpart of the admissibility-weight convergence argument of [4]. The reduction is exact only if $\epsilon(x)$ is defined as a quadratic occupancy in the rank-three sector $\mathrm{Sym}^2(V_\rho)$. It does not hold for an arbitrary additive scalar density.

Once this local weight reduction is established, the metric bridge follows from the principal-symbol reconstruction of L_{eff}:

$$g^{\tau\tau}(x) = -\frac{A_\tau(x)}{4}.$$

The factor $\frac{1}{4}$ is fixed by the LCII normalisation $R^2 = -2g^{\tau\tau}$, together with the flat-space values $A_\tau^\infty = 2$ and $g^{\tau\tau,\infty} = -\frac{1}{2}$. This identification belongs to the operator-symbol reconstruction of the effective metric [1, 3] and does not use the Einstein equation.

Section 2 defines the quadratic local occupancy precisely. Section 3 proves the local weight reduction (Lemma). Section 4 records the principal-symbol identification. Section 5 states and proves the main result (Theorem and Corollary). Section 6 discusses the role of the quadratic definition and the scope of the result.

2 Quadratic local occupancy

Let V_ρ be the Weil module of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ forced by admissibility [1]. The local rank-three effective sector is identified as

$$H_{\mathrm{eff}}(x) \simeq \mathrm{Sym}^2(V_\rho),$$

equipped with its SU(2)-invariant quadratic norm. The local Born–Infeld capacity is measured in this norm.

Definition 2.1 (Local quadratic occupancy)**.** Let $C_{\mathrm{occ}}(x)$ denote the occupied part of the local admissible sector over x, and let $C_{\max}(x)$ denote a maximally saturated admissible vector, with

$$\sigma_{\max} := \|C_{\max}(x)\|^2_{\mathrm{Sym}^2(V_\rho)} = c_\chi^2.$$

The local occupancy field is

$$\epsilon(x) := \frac{\|C_{\mathrm{occ}}(x)\|^2_{\mathrm{Sym}^2(V_\rho)}}{\sigma_{\max}}, \qquad 0 \le \epsilon(x) \le 1.$$

Equivalently, in any admissible orthonormal frame of $\mathrm{Sym}^2(V_\rho)$,

$$\epsilon(x) = \frac{\sum_{a=1}^{3} |c_a^{\mathrm{occ}}(x)|^2}{\sum_{a=1}^{3} |c_a^{\max}(x)|^2}.$$

This expression is frame-independent because the norm is the SU(2)-invariant quadratic norm on $\mathrm{Sym}^2(V_\rho)$. The field $\epsilon(x)$ is not introduced as an independent scalar density. It is the fraction of the local admissible fibre already occupied in the quadratic norm that defines the Born–Infeld capacity.

3 Local admissibility-weight reduction

Lemma 3.1 (Local admissibility-weight reduction)**.** *Let $A_\tau^\infty = 2$ be the homogeneous temporal admissibility coefficient. If a fraction $\epsilon(x)$ of the local $\mathrm{Sym}^2(V_\rho)$ capacity is occupied, then the temporal coefficient of the local effective admissibility operator satisfies*

$$A_\tau(x) = A_\tau^\infty\big(1 - \epsilon(x)\big) = 2\big(1 - \epsilon(x)\big).$$

Proof. The effective operator L_{eff} is local at the level of its principal symbol. At each point x, its temporal coefficient is the normalised trace of the admissibility weight operator Ad over the fibre sector available over x [4]. In the

homogeneous case, the full image sector is available, and the fibre-averaging argument of [4] gives $A_\tau^\infty = 2$.

Locally, the occupied and free parts of the fibre decompose orthogonally in the SU(2)-invariant quadratic norm on $\mathrm{Sym}^2(V_\rho)$:

$$C_{\max}(x) = C_{\mathrm{occ}}(x) \oplus C_{\mathrm{free}}(x).$$

Therefore $\|C_{\mathrm{free}}(x)\|^2 = \|C_{\max}(x)\|^2 - \|C_{\mathrm{occ}}(x)\|^2$, and dividing by $\sigma_{\max}$ gives

$$\frac{\|C_{\mathrm{free}}(x)\|^2}{\|C_{\max}(x)\|^2} = 1 - \epsilon(x).$$

Since $\mathrm{Sym}^2(V_\rho)$ is an irreducible SU(2)-representation and the admissibility weight Ad is SU(2)-equivariant, Schur's lemma forces $\mathrm{Ad} = \lambda \cdot \mathrm{Id}$ on this sector for some scalar $\lambda > 0$. Therefore the partial trace over any subspace $C \subseteq \mathrm{Sym}^2(V_\rho)$ satisfies

$$\mathrm{tr}_C(\mathrm{Ad}) = \lambda \|C\|^2.$$

Consequently,

$$\frac{\mathrm{tr}_{C_{\mathrm{free}}(x)}(\mathrm{Ad})}{\mathrm{tr}_{C_{\max}(x)}(\mathrm{Ad})} = \frac{\|C_{\mathrm{free}}(x)\|^2}{\|C_{\max}(x)\|^2} = 1 - \epsilon(x).$$

The local temporal coefficient is the normalised trace over $C_{\mathrm{free}}(x)$:

$$A_\tau(x) = A_\tau^\infty \frac{\mathrm{tr}_{C_{\mathrm{free}}(x)}(\mathrm{Ad})}{\mathrm{tr}_{C_{\max}(x)}(\mathrm{Ad})} = A_\tau^\infty \big(1 - \epsilon(x)\big).$$

Using $A_\tau^\infty = 2$ yields $A_\tau(x) = 2\big(1 - \epsilon(x)\big)$. □

4 Principal-symbol identification

The effective metric is reconstructed from the principal symbol of L_{eff} [1]. For the temporal component, in the LCII normalisation [2], this gives

$$g^{\tau\tau}(x) = -\frac{A_\tau(x)}{4}.$$

This identification follows from the principal-symbol formula of [1], applied locally, together with the flat-space normalisation $A_\tau^\infty = 2$, $g^{\tau\tau,\infty} = -\frac{1}{2}$ established in [3].

This step is purely operator-theoretic. It uses the local principal symbol of the admissibility operator and does not use the Einstein equation, the spectral-entropy variation, or the field equations of the Gravity paper [1].

5 Spectral closure of the capacity–metric bridge

Theorem 5.1 (Spectral closure of the local capacity–metric bridge)**.** *The local capacity radius and the metric radius coincide:*

$$R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x).$$

Equivalently,

$$\epsilon(x) = 1 + 2g^{\tau\tau}(x).$$

Proof. By Definition 2.1, $R(x)^2 = 1 - \epsilon(x)$. By Lemma 3.1, $A_\tau(x) = 2\big(1 - \epsilon(x)\big)$. The local principal-symbol identification of Section 4 gives $g^{\tau\tau}(x) = -A_\tau(x)/4$. Hence

$$-2g^{\tau\tau}(x) = \frac{A_\tau(x)}{2} = 1 - \epsilon(x).$$

Therefore $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$, and rearranging gives the equivalent form. □

Corollary 5.2 (Upgrade of Remark 3.2 of [2])**.** *The equivalence between the capacity and metric definitions of $R(x)$, postulated in Remark 3.2 of [2], is not an independent postulate. It follows from local admissibility-weight reduction on the quadratic fibre sector* $\mathrm{Sym}^2(V_\rho)$*. The chain*

$$\epsilon(x) \longmapsto A_\tau(x) \longmapsto g^{\tau\tau}(x)$$

is closed spectrally, without importing the Einstein equation.

Remark 5.3 (Role of the quadratic occupancy). The proof depends essentially on $\epsilon(x)$ being a quadratic occupancy in $\mathrm{Sym}^2(V_\rho)$. If $\epsilon(x)$ were only an additive scalar density, the Schur step would not apply, the orthogonal decomposition $C_{\max}(x) = C_{\mathrm{occ}}(x) \oplus C_{\mathrm{free}}(x)$ would not imply the exact factor $1 - \epsilon(x)$ in $A_\tau(x)$, and the bridge would remain open. The identity $\epsilon(x) = 1 + 2g^{\tau\tau}(x)$ is therefore a spectral admissibility identity, not a phenomenological identification between matter density and metric curvature.

6 Discussion

The result shows that $\epsilon(x)$ and $g^{\tau\tau}(x)$ are not independent fields: they are two readings of the same local admissibility datum. The projective capacity available at x for further temporal transitions is $1-\epsilon(x)$, and this is precisely the quantity that the effective metric encodes in its temporal component $g^{\tau\tau}(x) = -\frac{1}{2}(1 - \epsilon(x))$.

Gravitational time dilation then takes the following form in the capacity language. A massive configuration at x occupies a fraction $\epsilon(x) > 0$ of the local BI budget; the residual temporal capacity $R(x) = \sqrt{1-\epsilon(x)} < 1$ slows local clocks relative to the vacuum. This is not imported from the Einstein equation. It is a consequence of the reduction of the available admissible fibre sector.

The Schwarzschild case provides a consistency check via [2], Corollary 6.1: in the weak-field regime, $R(x) \approx 1 + \Phi_N(x)/c^2$, so $\epsilon(x) \approx -2\Phi_N(x)/c^2$, with $\Phi_N < 0$ near a mass source. The capacity budget is slightly more occupied near a massive body, reducing the temporal residual, consistently with the standard gravitational redshift.

7 Conclusion

We have closed LC-O2-O1. The local capacity radius $R(x)^2 = 1 - \epsilon(x)$ and the metric radius $R(x)^2 = -2g^{\tau\tau}(x)$ are the same quantity, once the occupancy field is defined as a quadratic occupancy in $\mathrm{Sym}^2(V_\rho)$.

The proof proceeds through two steps. First, the local temporal coefficient of the admissibility operator is reduced by the free quadratic capacity fraction:

$$A_\tau(x) = 2\big(1 - \epsilon(x)\big).$$

This follows from the SU(2)-invariance of the admissibility weight (Schur's lemma) and the orthogonality of the occupied/free decomposition. Second, the principal-symbol identification gives

$$g^{\tau\tau}(x) = -\frac{A_\tau(x)}{4}.$$

Combining these two identities yields $R(x)^2 = 1 - \epsilon(x) = -2g^{\tau\tau}(x)$. The bridge is closed spectrally, without the Einstein equation. The open bridge of Remark 3.2 of [2] is hereby upgraded to a theorem.

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